

Shayan Mohajer Hamidi

Department of Electrical Engineering, Stanford University, CA, USA

☎ 650-444-9763 • ✉ smohajer@stanford.edu
Google Scholar LinkedIn Website Github

Education

Stanford University:

- Postdoctoral Scholar 05/2025-current
- *Focus:* Post-training of LLMs, Reasoning in LLMs, Diffusion Models.

University of Toronto:

- Postdoctoral Scholar 01/2025-04/2025
- *Focus:* Diffusion Models.

University of Waterloo:

- Ph.D. in Electrical and Computer Engineering—Overall GPA: 91.6/100. 01/2020-01/2025
- M.A.Sc. in Electrical and Computer Engineering—Overall GPA: 90.2/100. 09/2016-09/2018

Sharif University of Technology:

- B.Sc. in Electrical and Computer Engineering—Overall GPA: 17.87/20. 09/2011-04/2016
- B.Sc. in Mathematics.

Research & Applied ML Expertise

- **Reasoning in Large Language Models:** Enhancing multi-step reasoning, planning, and structured thought processes in LLMs using instruction tuning and chain-of-thought guidance.
- **Autonomous Agents:** Advancing the capabilities of agentic AI systems by strengthening the reasoning and decision-making performance of underlying language model architectures.
- **Parameter-Efficient LLM Fine-Tuning:** Developing PEFT methods, including LoRA-based adaptation, to efficiently specialize large language models for domain-specific tasks.
- **Mathematics & Optimization:** Leveraging advanced mathematical and optimization techniques to analyze, model, and solve complex problems in machine learning and artificial intelligence.

Selected Publications

Conference Publications

1. **Hamidi, SM.**, Ye, L., Plataniotis, K.N.: “*DreamPhase: Offline Imagination and Uncertainty-Guided Planning for Large-Language-Model Agents*”. 2026 International Conference on Learning Representations (ICLR 2026).
2. **Hamidi, SM.***, Ye, L.*, Pilanci, M., Plataniotis, K.N.: “*CL-DPS: A Contrastive Learning Approach to Blind Nonlinear Inverse Problem Solving via Diffusion Posterior Sampling*”. Submitted to the 2026 International Conference on Learning Representations (ICLR 2026).
3. Ye, L., **Hamidi, SM.***, et. al. : “*ASMIT: Attention-Stabilized Multiple Instance Learning for Whole-Slide Imaging*”. Submitted to the 2026 International Conference on Learning Representations (ICLR 2026).
4. **Hamidi, SM.**, Yang, E.H., Liang, B.: “*Coupled Data and Measurement Space Dynamics for Enhanced Diffusion Posterior Sampling*”. 2025 The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025).
5. **Hamidi, SM.**, Yang, E.H.: “*Score-Based Manifold Projection for Diffusion-Based Inverse Problems*”. 2025 IEEE International Symposium on Information Theory (ISIT 2025).
6. **Hamidi, SM.**, Yang, E.H.: “*Conditional Mutual Information Based Diffusion Posterior Sampling for Solving Inverse Problems*”. 2025 IEEE International Symposium on Information Theory (ISIT 2025).
7. Yang, E.H., **Hamidi, SM.**: “*Coded Deep Learning: Framework and Preliminary Results*”. 2025 IEEE

International Symposium on Information Theory ([ISIT 2025](#)).

8. **Hamidi, SM.**, Berehyi, A.: “Rate-Constrained Quantization for Communication-Efficient Federated Learning”. 2025 IEEE International Conference on Acoustics, Speech and Signal Processing ([ICASSP 2025](#)).
9. **Hamidi, SM.**, Berehyi, A.: “Second-Order Wireless Federated Learning via Nonparametric Hessian Estimation”. 2025 IEEE International Conference on Acoustics, Speech and Signal Processing ([ICASSP 2025](#)).
10. **Hamidi, SM.***, Ye, L.*, Tan, R. *, Yang, E.H.: “Bayes Conditional Distribution Estimation for Knowledge Distillation Based on Conditional Mutual Information”. In 2024 International Conference on Learning Representations ([ICLR 2024](#)).
11. **Hamidi, SM.**, Deng, X., Tan, R. , Ye, L., Salamah, A.: “How to Train the Teacher Model for Effective Knowledge Distillation”. European Conference on Computer Vision, ([ECCV 2024](#)).
12. **Hamidi, SM.**, Ye, L., Tan, R. , Yang, E.H.: “Robustness Against Adversarial Attacks via Learning Confined Adversarial Polytopes”. 2024 IEEE International Conference on Acoustics, Speech and Signal Processing ([ICASSP 2024](#)).
13. **Hamidi, SM.**, et al.: “Fed-IT: Addressing Class Imbalance in Federated Learning through an Information-Theoretic Lens”. In 2024 IEEE International Symposium on Information Theory ([ISIT 2024](#)).
14. Yang, E.H., **Hamidi, SM.**, Ye, L., Tan, R. , Yang, B.: “Conditional Mutual Information Constrained Deep Learning: Framework and Preliminary Results”. In 2024 IEEE International Symposium on Information Theory ([ISIT 2024](#)).
15. **Hamidi, SM.**, Mehrabi, M., Khandani, A. K., Gündüz, D.: “Over-the-Air Federated Learning Exploiting Channel Perturbation”. In 2022 IEEE 23rd International Workshop on Signal Processing Advances in Wireless Communication ([SPAWC 2022](#)) (pp. 1-5). IEEE.

Journal Publications.....

1. Yang, E.H. and **Hamidi, SM.**: “Coded Deep Learning: Framework and Algorithm” IEEE Transactions on Information Theory ([TIT 2025](#)).
2. **Hamidi, SM.**, Berehyi, A., Asaad, S., Poor, H.V.: “Over-the-Air Fair Federated Learning via Multi-Objective Optimization” IEEE Communications Letters ([Comm Letters 2025](#)).
3. Yang, E.H., **Hamidi, SM.**, L Ye, R Tan, B Yang. “Conditional Mutual Information Constrained Deep Learning for Classification” IEEE Transactions on Neural Networks and Learning Systems ([TNNLS 2025](#)).
4. **Hamidi, SM.**, Ye., L.: “Distributed Quasi-Newton Method for Fair and Fast Federated Learning” Transactions on Machine Learning Research ([TMLR 2025](#)).
5. Ye, L., **Hamidi, SM.**, Yang, E.H.: “Towards undistillable models by minimizing conditional mutual information” Transactions on Machine Learning Research ([TMLR 2025](#)).
6. Salamah, AH., **Hamidi SM.**, Yang, E.H.: “A coded knowledge distillation framework for image classification based on adaptive JPEG encoding.” ([Pattern Recognition 2025](#)).
7. **Hamidi, SM.**, Yang, E.H.: “AdaFed: Fair Federated Learning via Adaptive Common Descent Direction.” Transactions on Machine Learning Research ([TMLR 2024](#)).
8. **Hamidi, SM.**, Damen, O.: Fair Wireless Federated Learning through the Identification of a Common Descent Direction, IEEE Communications Letters ([Comm Letters 2024](#)).
9. **Hamidi, SM.**: Training Neural Networks on Remote Edge Devices for Unseen Class Classification, IEEE Signal Processing Letters ([LSP 2024](#)).
10. **Hamidi, SM.**, et al.: “A Learning-Based Framework for Locating Faults on Power Grid Lines Based on Distortion of Traveling Waves.” IEEE Transactions on Instrumentation and Measurement ([TIM 2022](#)).

Patents.....

1. Yang, E.H., **SM Hamidi**: “Methods and systems for conditional mutual information constrained deep learning”, [US Patent](#) App. 18/829,437, 2024.
2. Yang, E.H., **SM Hamidi**. “Systems and methods for training deep learning models”, [US Patent](#) App.

18/666,958, 2024.

3. **SM Hamidi**, Sanjeewa Herath, Alireza Bayesteh, Amir Keyvan Khandani. “*Systems and methods for communication resource usage control*”, [US Patent](#) App. 15/824,352, 2019.
4. P Struhsaker, P Posner, **SM Hamidi**, AK Khandani. “*Methods for formation of antenna array from sub-arrays*”, [US Patent](#) 10,756,443, 2020.

Preprints.....

1. **Hamidi, SM.**, Yang, E.H.: “*Enhancing diffusion models for inverse problems with covariance-aware posterior sampling*”, arXiv preprint arXiv:2412.20045.

Core Machine Learning Expertise

ML Frameworks: PyTorch, Hugging Face (Transformers, Datasets), Scikit-learn

ML/LLM Tools: LLM Fine-tuning (LoRA, QLoRA), RLHF (trl)

MLOps/Cloud: Linux, Docker, Google Cloud, MLOps concepts

Programming Languages: Python, R, MATLAB

Work Experience

2018–2020 Research Associate, University of Waterloo.....

Data Analyst in the Multimedia Lab, supervised by Prof. Amir Khandani. Analyzed QoS metrics (latency, jitter, packet loss) in streaming datasets, identified usage patterns related to congestion and device behavior, and developed models to anticipate degradation and inform network optimization.

Honors and Awards

2007: **Bronze Medal**, 17th Iranian National **Informatics Olympiad**, Iran.

2008: **Silver Medal**, 26th Iranian National **Mathematical Olympiad**, Iran.

2011: Ranked 61th Among More Than 300,000 Participants in the Nationwide University Entrance Examination Known as Konkoor for Entering B.Sc Degree.

2017: Winner of the Winter 2017 TA awards for being Tutorial TA for ECE316, University of Waterloo.

2017: Faculty of Engineers Awards for being among top-GPA M.A.Sc. students, University of Waterloo.

2022: Prof. Jon W. Mark Graduate Scholarship in Communication, University of Waterloo, Canada.

2025: *J2C Certification Award* for TMLR Submission “Distributed Quasi-Newton Method for Fair and Fast Federated Learning”.

Executive Experience

Conference Session Chair.....

- 2025 IEEE International Symposium on Information Theory (ISIT 2025)

Reviewing Services.....

- The International Conference on Machine Learning (ICML)
- The International Conference on Learning Representations (ICLR)
- Annual Conference on Neural Information Processing Systems (NeurIPS)
- Conference on Empirical Methods in Natural Language Processing (EMNLP)
- The European Conference on Computer Vision (ECCV)
- IEEE Transactions on Information Theory
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Wireless Communications
- IEEE Transactions on Vehicular Technology
- IEEE Communication Letters
- Transactions on Machine Learning Research (TMLR)

- International Conference on Computer Vision (ICCV)
- IEEE International Symposium on Information Theory (ISIT)
- The British Machine Vision Conference (BMVC)
- Pattern Recognition

Teaching Assistant

Spring 2023: Probability and Statistics 1 (ECE203), University of Waterloo.

Winter 2023: Digital Communications (ECE318), University of Waterloo.

Fall 2023: Probability and Statistics 2 (ECE307), University of Waterloo.

Spring 2022: Probability and Statistics 1 (ECE203), University of Waterloo.

Winter 2021: Probability and Statistics 2 (ECE307), University of Waterloo (*received best TA award*).

Mentorship

2022-2024: Ye, L. - Master's Thesis: Conditional Mutual Information in Knowledge Distillation.

2022-2024: Cameron Tan - Master's Thesis: Robust Conditional Mutual Information in Adversarial Training.

2024: Likhil Babbu - Master's Course Project: Uniformity in Training of Deep Neural Networks.

References

Mert Pilanci

Assistant Professor at Stanford University

Email: pilanci@stanford.edu

Ben Liang

Full Professor at the University of Toronto

Email: liang@ece.utoronto.ca

Yang, E.H.

Full Professor at the University of Waterloo

Email: ehyang@uwaterloo.ca